

# Penguin Wranglin'

Name: \_\_\_\_\_ Group members: \_\_\_\_\_

## Introduction

Our goal for today is to revisit the verbs of **dplyr** and consider how they can be used to answer statistical questions with data. In particular, we are going to focus on the **variables, verbs, and order** needed to answer each question, using pen and paper only. Each member of your group should keep a record of this document, as we will be implementing our solutions in R on Wednesday.

To be clear: you do not need to use full **dplyr** syntax today. An answer like:

Filter for college enrollment less than 10,000 → Group by region → summarize median debt

Is an entirely valid way to answer (underlining verbs and putting boxes around variables (or something) is not required, but could be helpful)

## The data

Each row describes one measured penguin. This curated dataset contains 333 observations and 8 variables, with no missing values. Measurements were recorded in 2007, 2008, and 2009.

| Variable          | Meaning                                                   |
|-------------------|-----------------------------------------------------------|
| species           | Adelie, Chinstrap, or Gentoo                              |
| island            | Biscoe, Dream, or Torgersen                               |
| bill_length_mm    | Bill length in millimeters                                |
| bill_depth_mm     | Bill depth in millimeters                                 |
| flipper_length_mm | Flipper length in millimeters                             |
| body_mass_g       | Body mass in grams                                        |
| sex               | Female or male (stored as <b>female</b> and <b>male</b> ) |
| year              | Year of measurement                                       |

## Verb reference

| Verb             | Purpose                                                |
|------------------|--------------------------------------------------------|
| <b>select</b>    | Keep specified columns.                                |
| <b>filter</b>    | Keep rows meeting a condition.                         |
| <b>mutate</b>    | Create or change variables; ordinarily preserves rows. |
| <b>group_by</b>  | Establish groups; does not itself collapse rows.       |
| <b>summarize</b> | Calculate summaries: here, one row per group.          |
| <b>arrange</b>   | Put rows in a specified order.                         |

Counts can be calculated in **summarize**; **count** is also valid. **Not every plan needs every verb.** For plots, name the plot, variables, and what its marks represent; no **ggplot2** syntax is needed.

Finally, for each problem you will need to decide on a *name* for the resulting dataset. **college\_region\_summary** is a good name, **college2** and **q1part2** are not. After you indicate each set of operations, I will ask you questions about the resulting dataset.

## Problem 1

For each species, describe the average body mass and its standard deviation. Include how many penguins contribute to each summary.

**Variables and plan:** Give the sequence of variables and verbs to address this problem. What are you calling the resulting dataset?

**Output and meaning:** What does one row represent in the final data? Which columns remain? Describe what you could learn from this dataset.

**Plot:** What graph would compare body-mass distributions? Would it use individual observations or the summary table?

## Problem 2

Among female penguins measured in 2008, how does average flipper length differ across species?

**Variables and plan:** Give the sequence of variables and verbs to address this problem. What are you calling the resulting dataset?

**Output and meaning:** What does one row represent in the final data? Which columns remain? Describe what you could learn from this dataset.

### Problem 3

Start with all penguins. Create body mass in kilograms; group by species and sex; calculate mean mass in kilograms; arrange from largest to smallest mean.

**Variables and plan:** Give the sequence of variables and verbs to address this problem. What are you calling the resulting dataset?

**Interpret:** State the research question, resulting columns, and meaning of the first row. Does it identify the heaviest individual penguin?

### Problem 4

**Plan A:** Retain penguins weighing at least 4,000 g, then calculate mean body mass by species.

**Plan B:** Calculate mean body mass by species, then retain species with means of at least 4,000 g.

**Identify:** Do these answer the same question? Explain what each answers.

**Compare:** Identify the variables, verbs, and the condition used in each plan. Explain what one output row represents in each.

**Discuss:** Must the same species appear in both outputs? Explain without calculating.

## Problem 5

For this exercise, label penguins weighing at least 4,000 grams “Big” and the rest “Small”. Does the proportion in the heavier category differ across species?

**Comparison:** First, be explicit. When determining this proportion, which penguins belong in the numerator and which in the denominator? Which plot best illustrates this comparison?

**Variables and plan:** Give the sequence of variables and verbs to address this problem. What are you calling the resulting dataset?

**Output and meaning:** What does one row represent in the final data? Which columns remain? Describe what you could learn from this dataset.

## Problem 6

A student wants to investigate the relationship between flipper length and body mass within each species. They first reduce the data to one mean flipper length and one mean body mass per species, then plot those three points. Does this answer their question? Revise the plan.

**Diagnose and revise:** What information was lost? Which variables and operations do you actually need?

**Plot:** Specify the axes, how to distinguish species, and what each point represents.